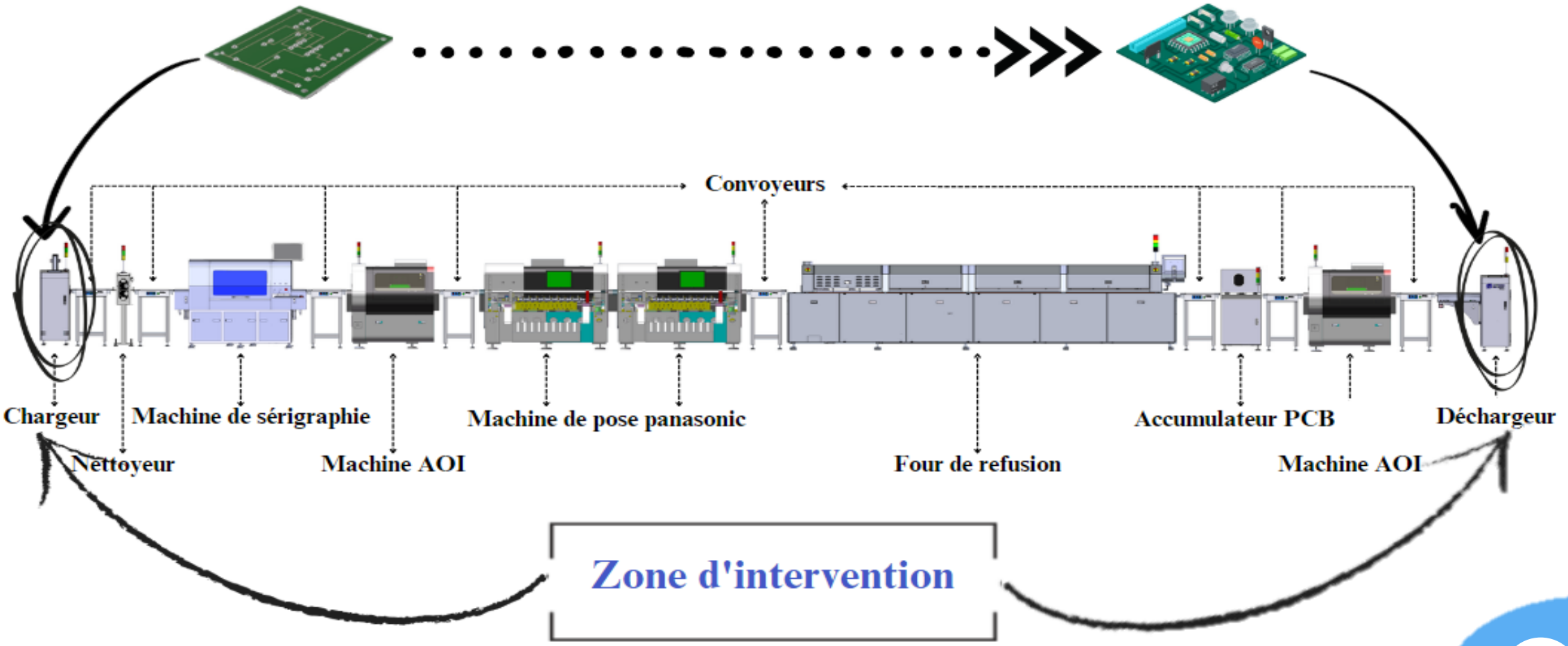
© Ahmed Hamdi

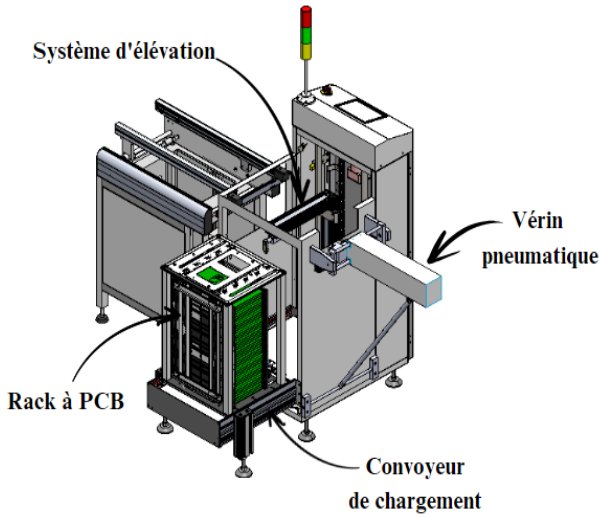
Processus de la ligne CMS



© Ahmed Hamdi

Mise en situation: Chargeur de ligne

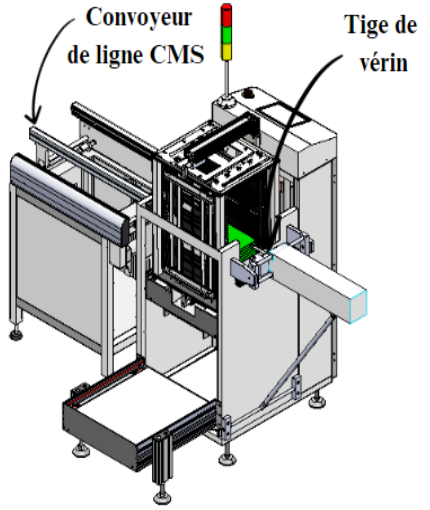
Etape 1



Placement et acheminement

- L'opérateur place le rack sur le convoyeur.
- Le convoyeur amène le rack vers l'élévateur.

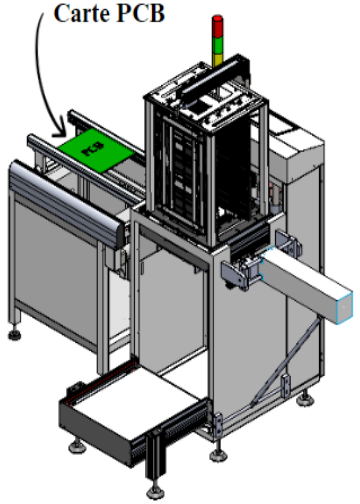
Etape 2



Transfert des cartes PCB

- Les cartes sont transférées une à une sur le convoyeur de la chaîne.

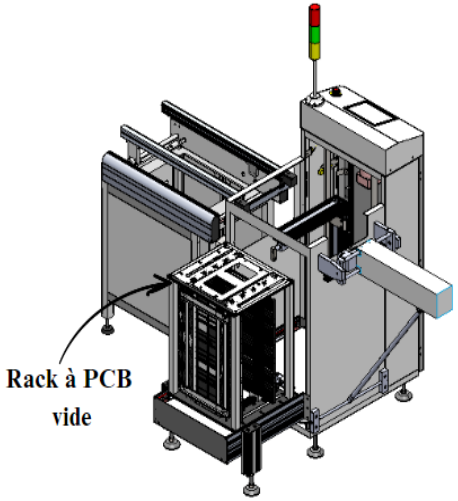
Etape 3



Descente de l'élévateur et évacuation

- L'élévateur redescend. Le rack vide est évacué.

Etape 4



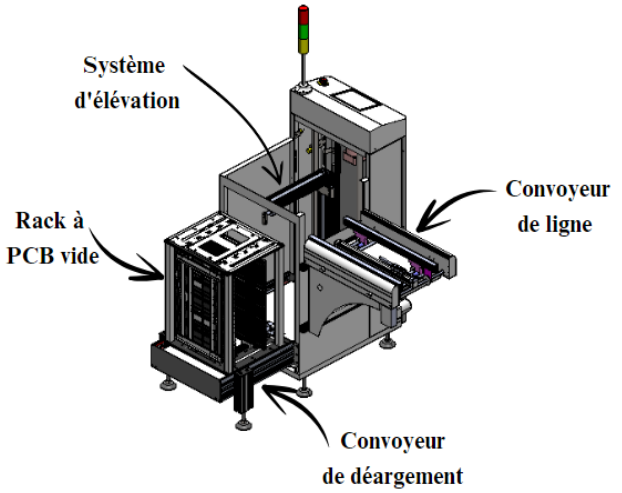
Recharge et reprise du cycle

- L'opérateur retire le rack vide et charge un nouveau rack.

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Mise en situation: Déchargeur de ligne

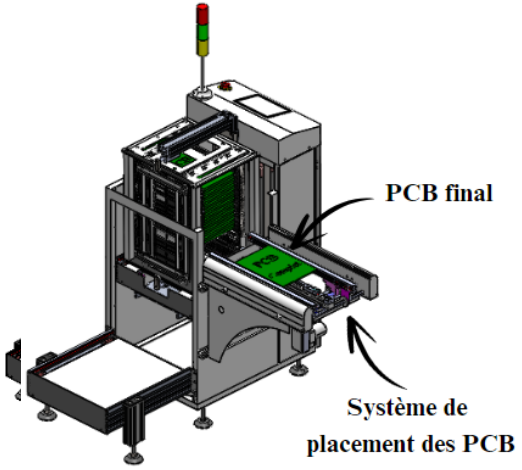
Etape 1



Placement et positionnement du rack vide

- L'opérateur place un rack vide sur le convoyeur.
- Le convoyeur achemine le rack vers l'élévateur.

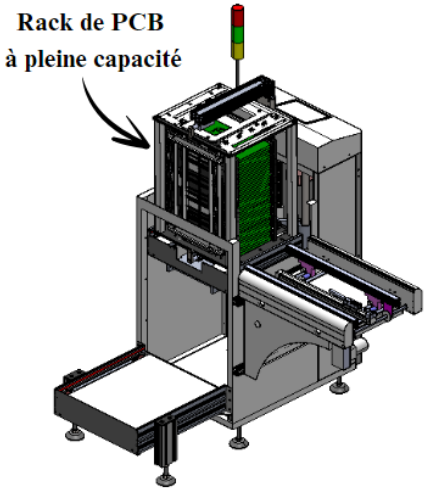
Etape 2



Transfert des cartes dans le rack

- Les cartes PCB sont transférées dans le rack une par une.

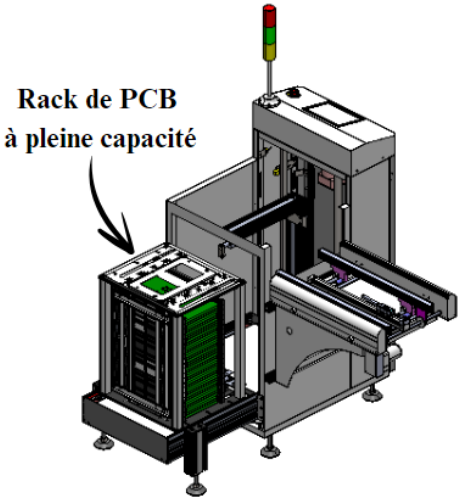
Etape 3



Descente de l'élévateur et évacuation

- L'élévateur redescend.
- Le rack rempli est évacué.

Etape 4

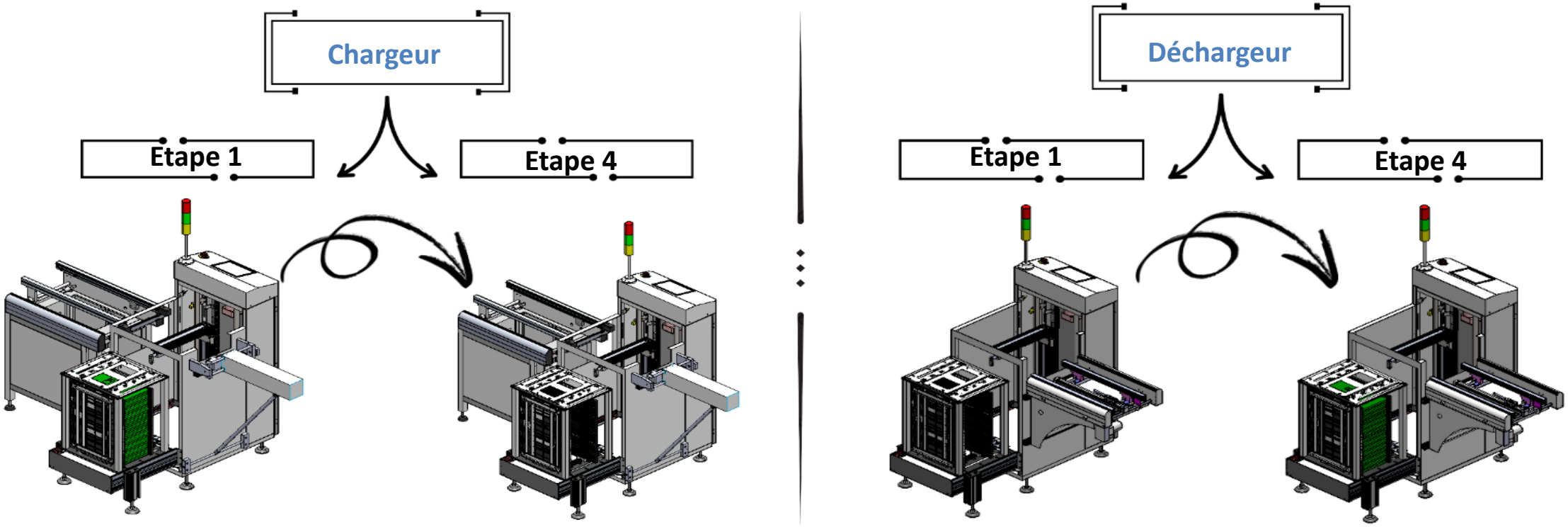


Recharge et reprise du cycle

- L'opérateur retire le rack rempli et place un nouveau rack vide.

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Problématique



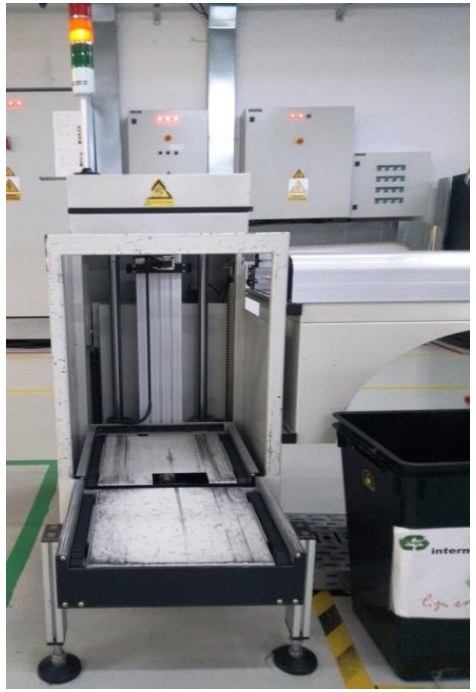
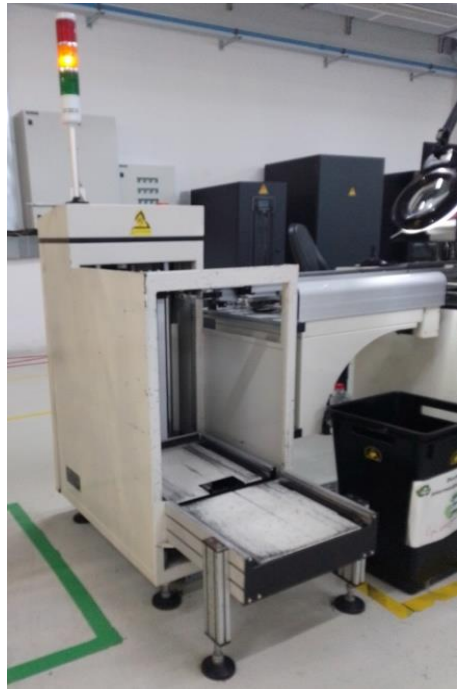
IMPORTANT

- Les machines de chargement et de déchargement ne peuvent contenir qu'un seul rack de PCB à la fois.
- Nécessité d'interventions répétées pour insérer et retirer les racks.

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Problématique

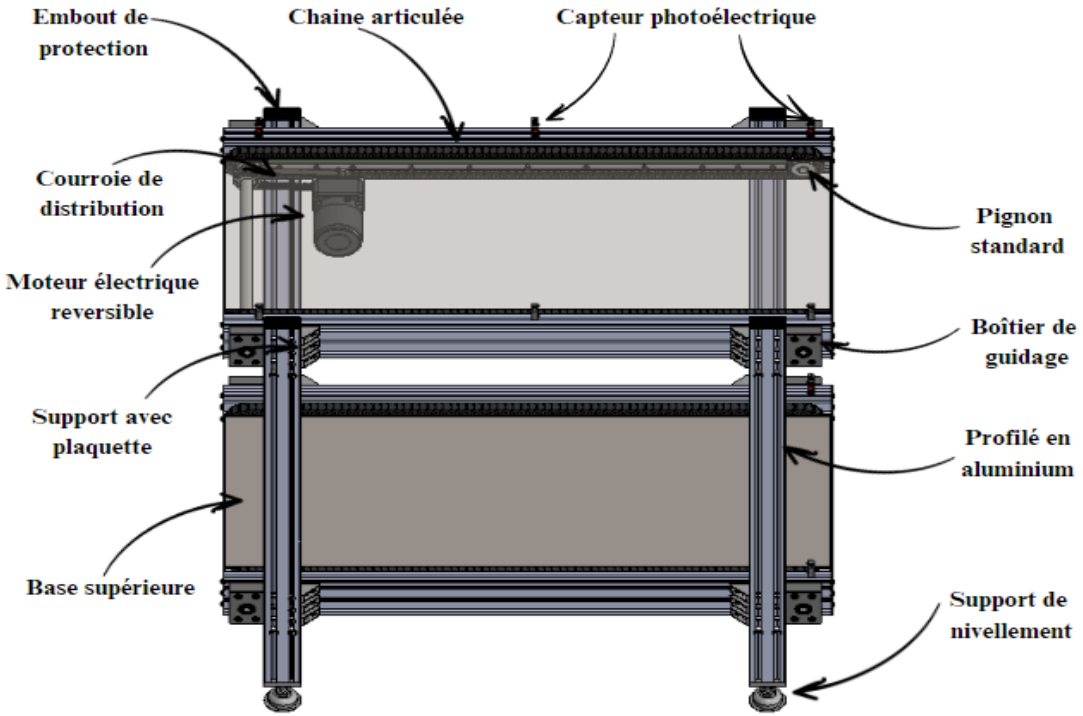


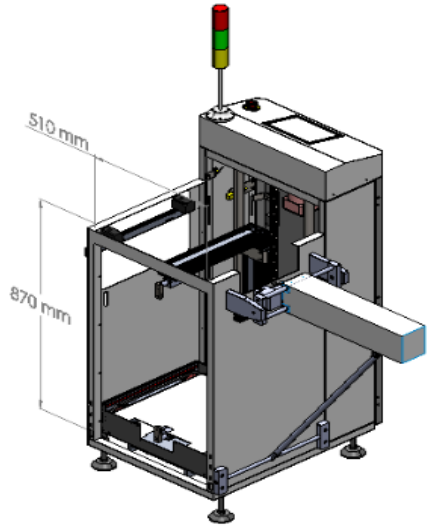



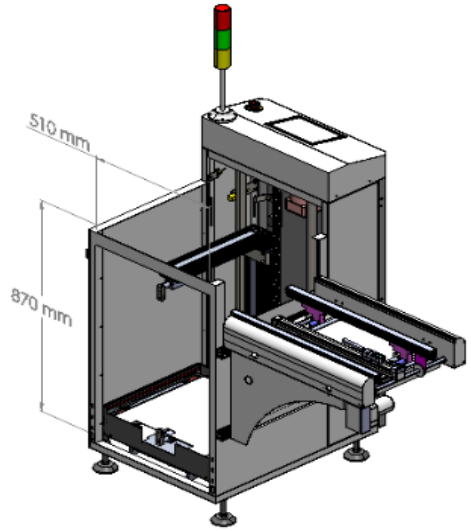
- **Interruptions régulières du flux de production, ralentissant l'avancement de la ligne CMS.**
- **Augmentation du temps de cycle, liée aux arrêts nécessaires aux opérations de manutention.**
- **Diminution de la productivité globale, due aux pertes de temps cumulées.**

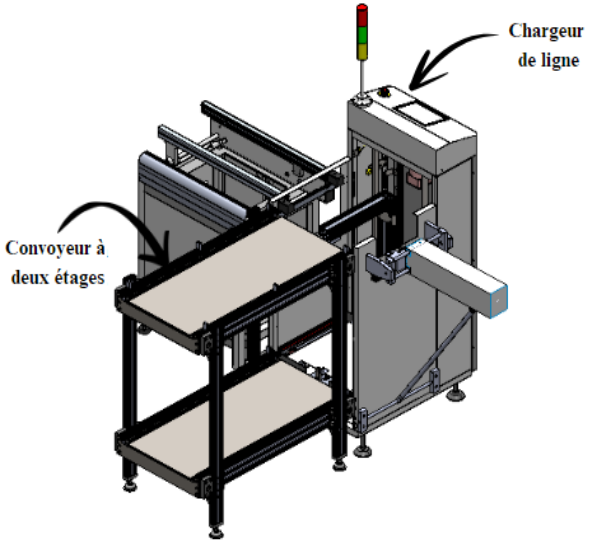
© Ahmed Hamdi

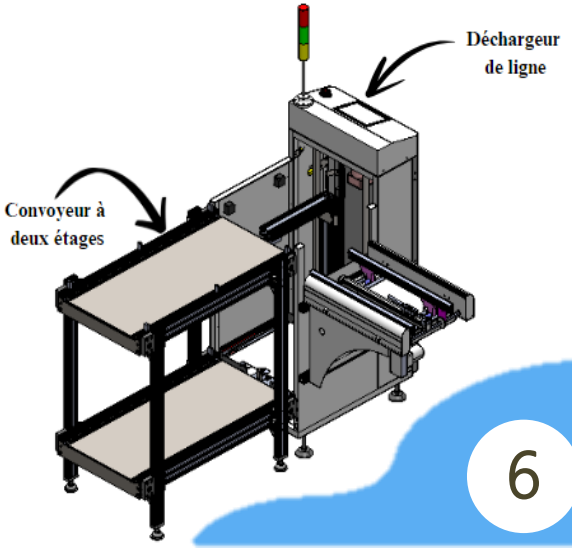
Solution proposée :Convoyeur à deux étages









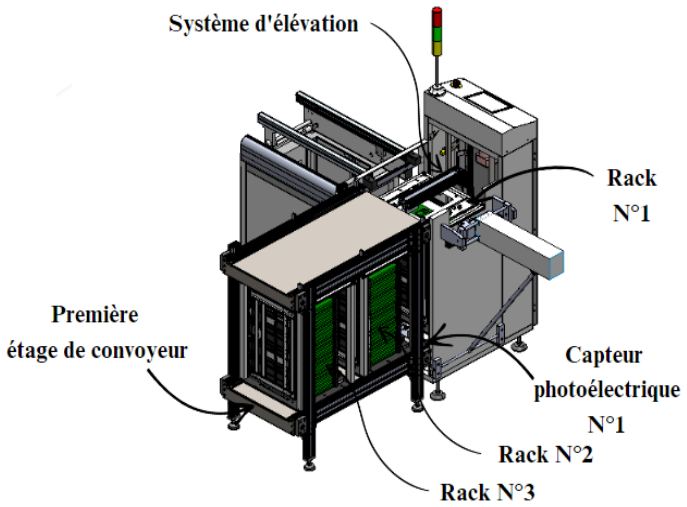


- Optimisation du flux de production
- Réduction des interventions manuelles
- Amélioration de l'efficacité globale
- Augmentation de la productivité
- Meilleure gestion des racks

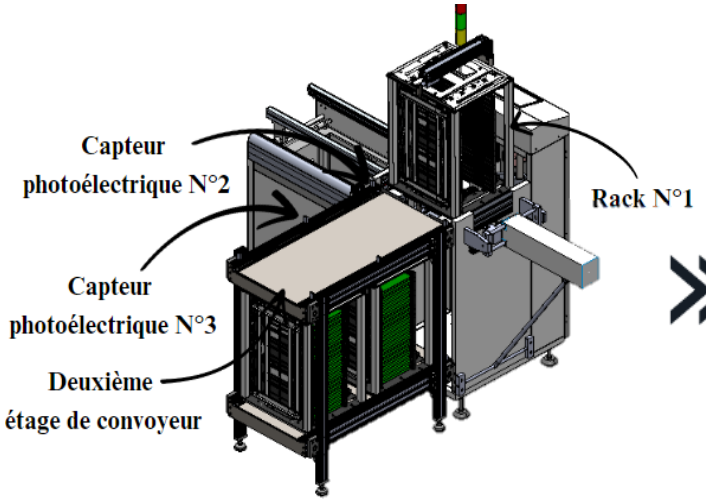
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Principe de fonctionnement : Chargeur de ligne

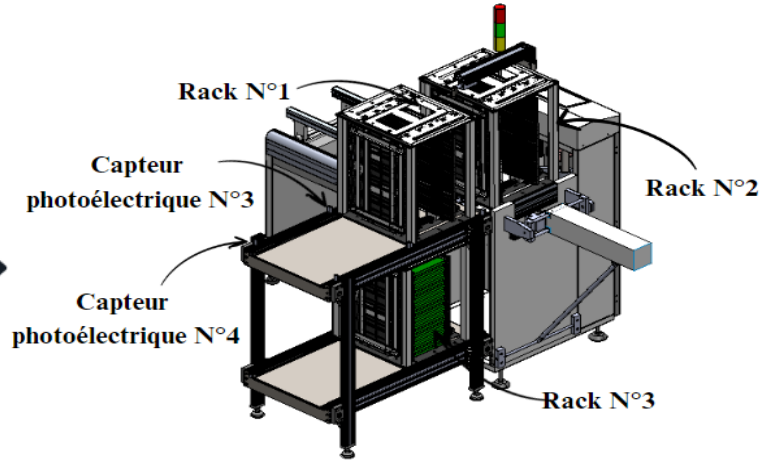
Chargement des racks



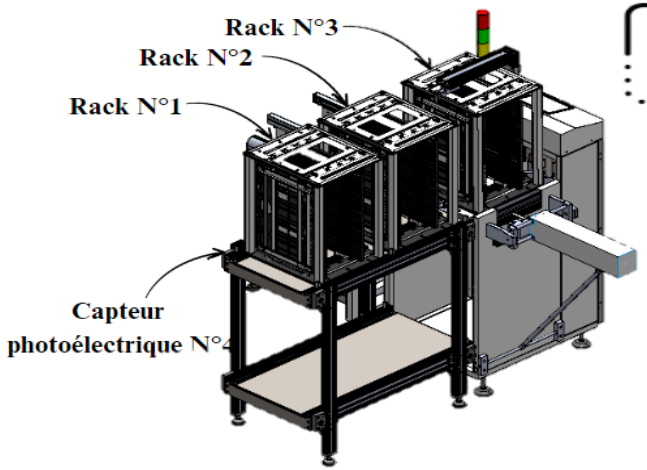
Détection et transfert le rack N°1



Détection et transfert le rack N°2



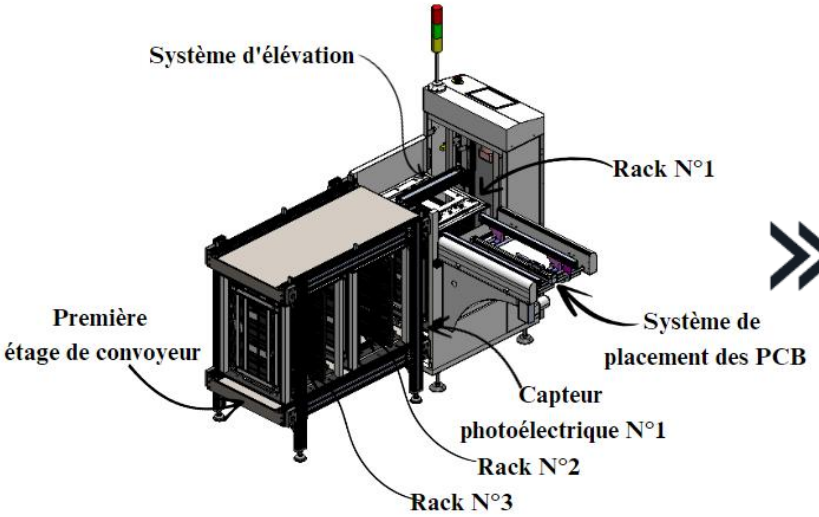
Détection et transfert le rack N°3



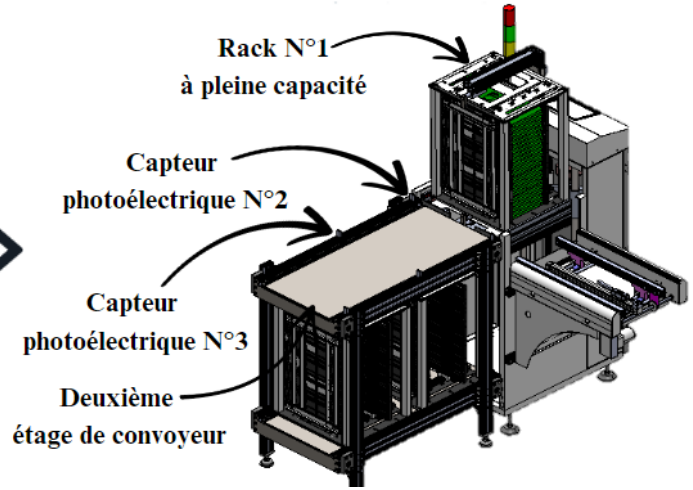
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Principe de fonctionnement : Déchargeur de ligne

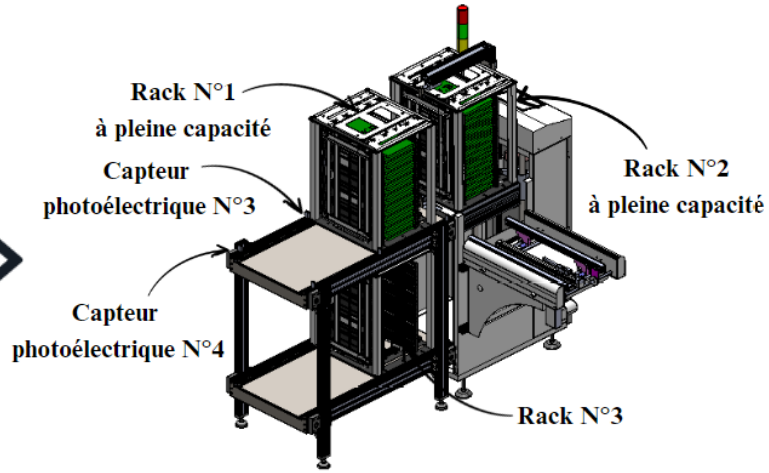
Chargement des racks



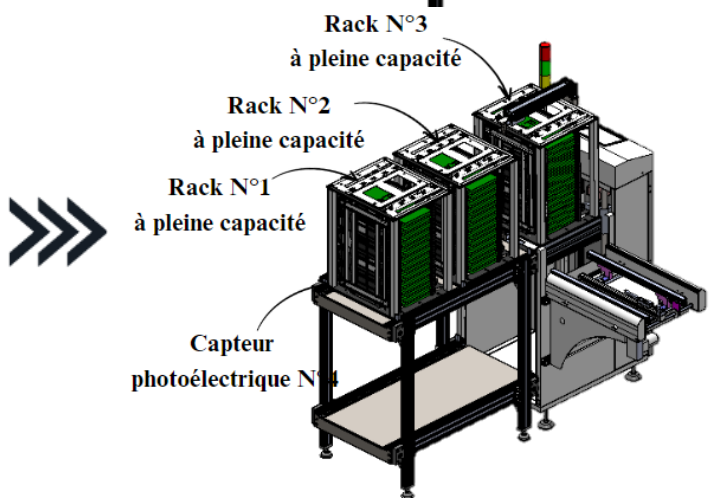
Détection et transfert le rack N°1

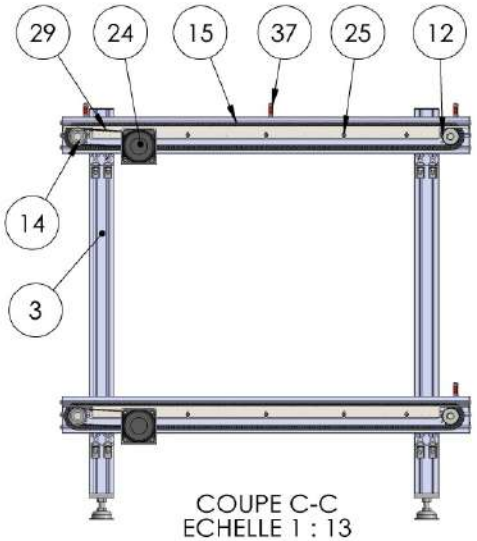


Détection et transfert le rack N°2

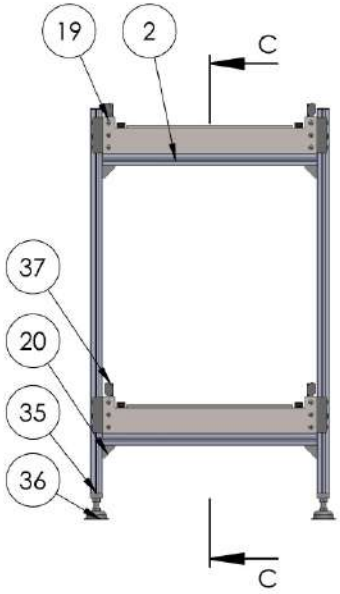


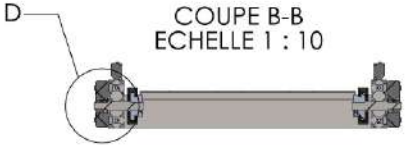
Détection et transfert le rack N°3



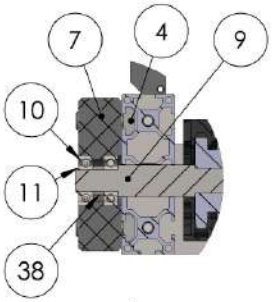


COUPE C-C
ECHELLE 1 : 13

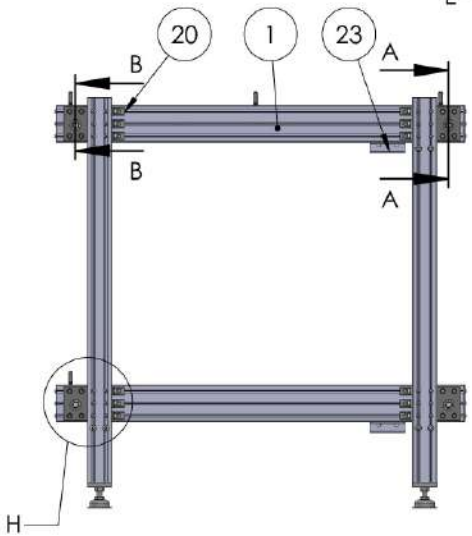


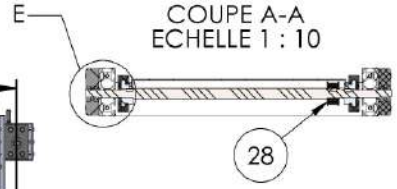


COUPE B-B
ECHELLE 1 : 10

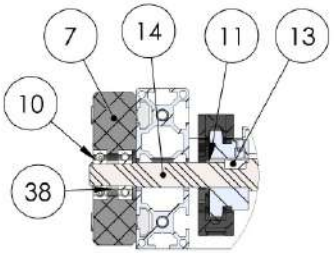


DÉTAIL D
ECHELLE 1 : 3

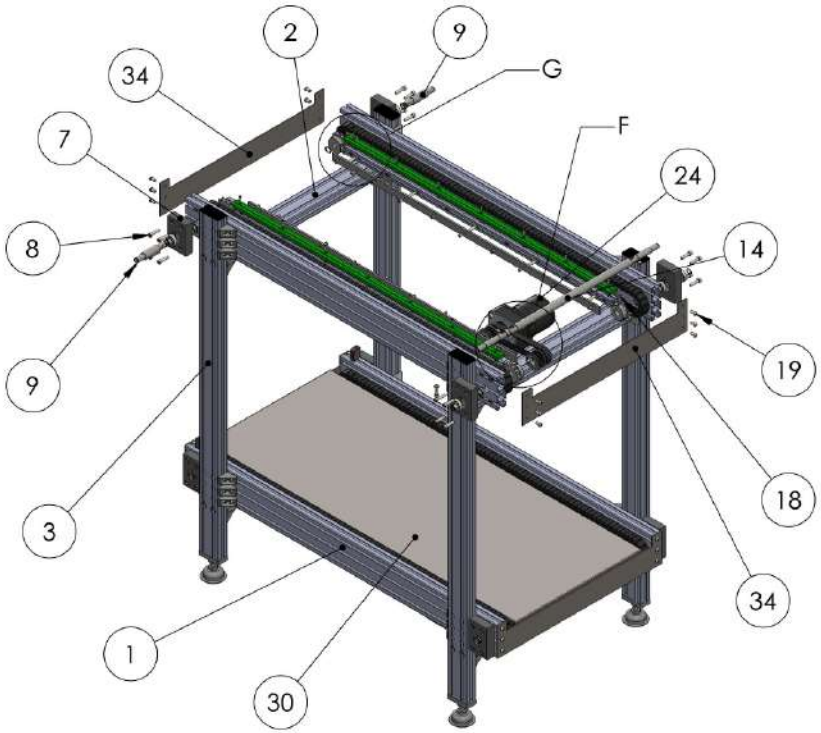


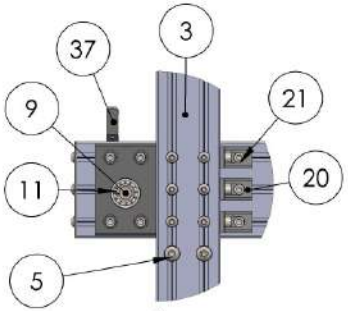


COUPE A-A
ECHELLE 1 : 10

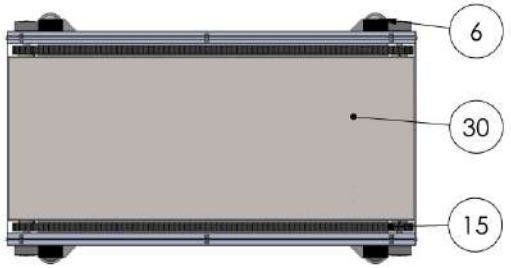


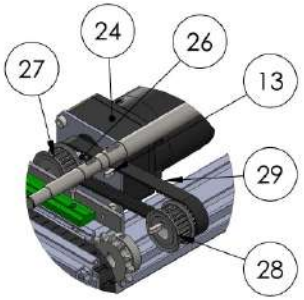
DÉTAIL E
ECHELLE 1 : 3



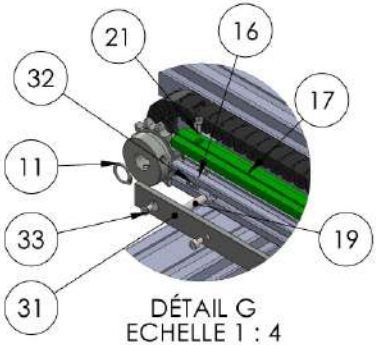


DÉTAIL H
ECHELLE 1 : 5



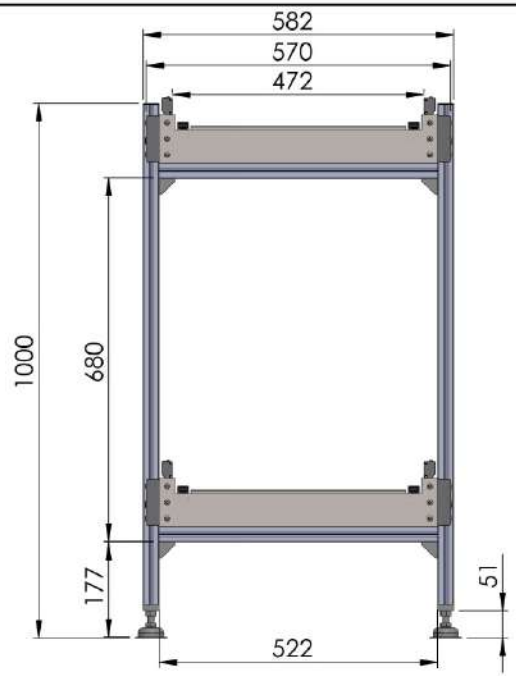


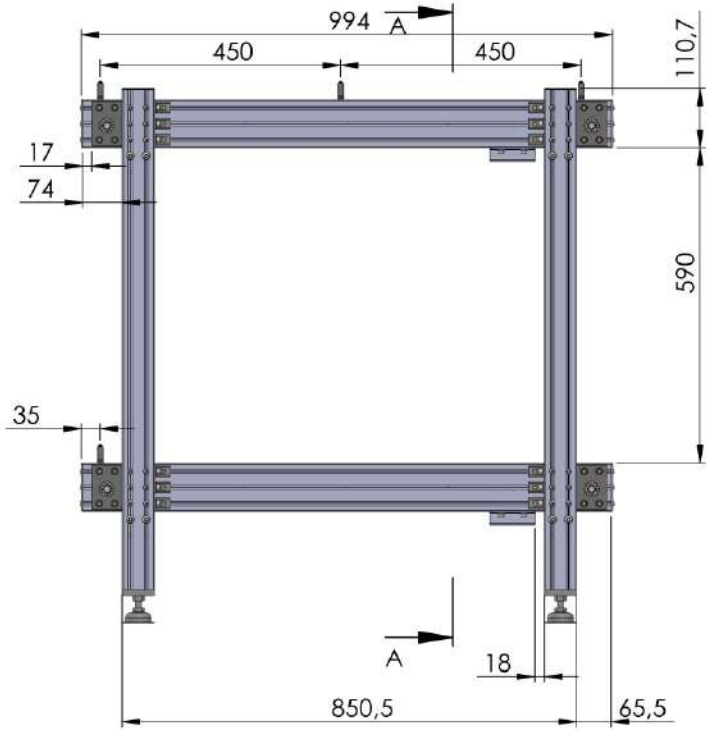
DÉTAIL F
ECHELLE 1 : 5

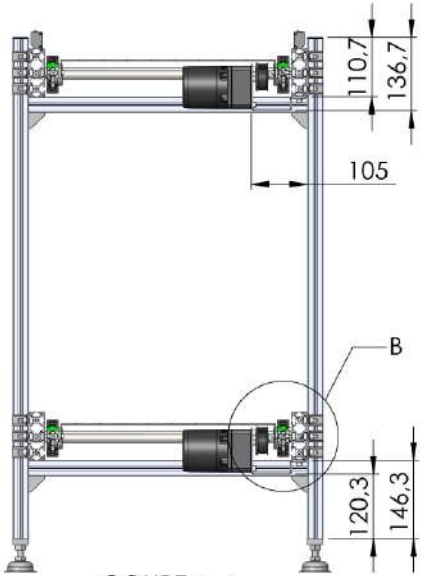


DÉTAIL G
ECHELLE 1 : 4

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Echelle : 1:13	**Convoyeur à deux étages**		Document 1
			Dessiné par: Hamdi.A
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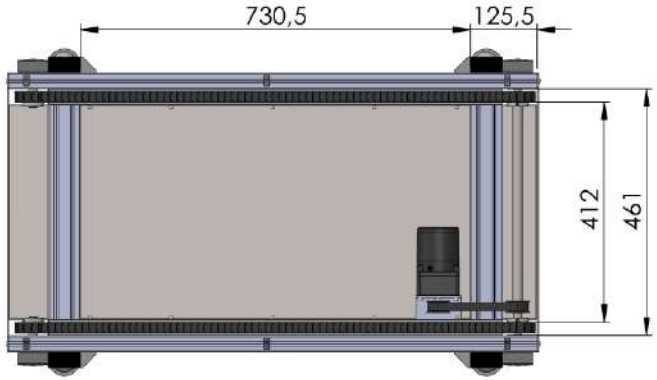


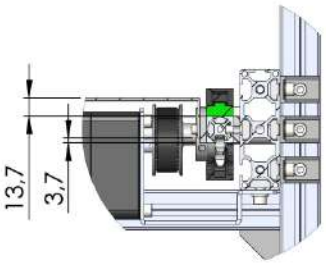




COUPE A-A
ECHELLE 1 : 10







DÉTAIL B
ECHELLE 1 : 4

Ecole Nationale d'Ingénieurs de Bizerte			
Echelle : 1:10	**Convoyeur à deux étages**		Document 1
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